Project Assignment

Retail Customer Segmentation and Campaign Response Prediction

1. Problem Statement

A retail company wants to understand its customers better. The company has collected customer profile, website activity, purchase behavior, and campaign response data. Your task is to analyze the dataset, identify meaningful customer groups, predict customer behavior, and provide simple business recommendations.

2. Dataset Description

Use the provided dataset file: retail_customer_project_dataset.csv

The dataset contains 600 customer records with the following columns:

Column(s)	Meaning
CustomerID	Unique customer identifier
Age, Gender	Basic demographic information
AnnualIncome, SpendingScore	Customer income and spending behavior
WebsiteVisitsPerMonth, AvgSessionMinutes	Online activity features
PreviousPurchases, PurchaseAmount	Buying history
DiscountUsed, EmailOpened	Marketing engagement indicators
MembershipYears	Relationship duration with the company
CampaignResponse	Whether the customer responded to the campaign
HighValueCustomer	Whether the customer is high value

3. Colab Setup Instructions

1. Open Google Colab and create a new notebook.
2. Upload retail_customer_project_dataset.csv using the Files panel or files.upload().
3. Import Pandas, NumPy, Matplotlib, and required scikit-learn libraries.
4. Read the CSV file and display the first few rows before starting the analysis.

```
from google.colab import files
uploaded = files.upload()

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df = pd.read_csv('retail_customer_project_dataset.csv')
df.head()
```

4. Tasks

Part A: Data Loading and Exploration

5. Load the dataset and display the first 10 rows.
6. Print dataset shape, column names, data types, and missing value count.
7. Find summary statistics for numerical columns.
8. Count customers by Gender, CampaignResponse, and HighValueCustomer.
9. Identify any unusual values or outliers using simple plots.

Part B: Data Preprocessing

10. Separate numerical and categorical columns.

11. Encode categorical columns such as Gender, DiscountUsed, EmailOpened, CampaignResponse, and HighValueCustomer.
12. Scale numerical features using StandardScaler.
13. Explain why encoding and scaling are required before clustering and classification.

Part C: Exploratory Data Analysis and Visualization

14. Plot histograms for Age, AnnualIncome, SpendingScore, and PurchaseAmount.
15. Create a scatter plot of AnnualIncome vs SpendingScore and color it by CampaignResponse.
16. Create a bar chart for CampaignResponse counts.
17. Create a correlation heatmap and identify at least three useful relationships.
18. Write 5 observations from your plots.

Part D: Customer Segmentation Using Clustering

19. Apply K-Means clustering using selected customer behavior features.
20. Use the elbow method for k = 1 to 10 and identify a suitable k.
21. Calculate silhouette score for the selected k.
22. Apply one additional clustering method: Agglomerative Clustering or DBSCAN.
23. Compare the clustering methods and describe the customer segments in simple words.

Part E: PCA and t-SNE Visualization

24. Apply PCA and reduce the scaled data to 2 components.
25. Plot clusters using PC1 and PC2.
26. Print the explained variance ratio.
27. Apply t-SNE and compare the visualization with PCA.
28. Explain which visualization is easier to interpret and why.

Part F: Supervised Learning

29. Choose one target: CampaignResponse or HighValueCustomer.
30. Split the data into training and testing sets using 80:20 split.
31. Train at least three models: Logistic Regression, Decision Tree, and KNN or Random Forest.
32. Evaluate models using accuracy, confusion matrix, precision, recall, and F1-score.
33. Create a comparison table and identify the best model.

Part G: Prompt Engineering and Business Recommendations

Create 5 prompts to explain the customer segments and campaign response results. Include the following prompt types:

- Basic prompt
- Role-based prompt
- Step-by-step prompt
- Data-driven prompt using your clustering statistics
- Business recommendation prompt

For each prompt, record the prompt, response summary, quality rating out of 5, and how you improved the prompt.

5. Questions

34. Which features are most useful for identifying high-value customers?
35. Do customers who opened emails have a higher campaign response rate?
36. Are high-income customers always high-spending customers? Support your answer using plots.
37. Which customer segment should the company target first and why?
38. What are the limitations of your model and clustering result?